

Emmanuel Idoko

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EDUCATION

University of Lagos

Bachelor of Science in Computer Science | CGPA: 4.68/5.0

Lagos, Nigeria

Expected 2027

- **Relevant Coursework:** Discrete Mathematics, Data Structures, Computational Methods, Linear Algebra, Calculus, Python/C Programming, Algorithm and Complexity, Compiler Construction, Concurrent Programming, Operating Systems, Distributed Systems, Machine Assembly Language.

Udacity

AI Programming with Python Nanodegree

Remote

Oct 2022 – Jan 2024

EXPERIENCE

AidCare

Founder & AI/ML Engineer

Lagos, Nigeria (Remote)

May 2025 – Present

- Designed and deployed a full-stack clinical AI platform on DigitalOcean with a containerized FastAPI backend, RBAC, scoped tokens, and anonymized logging — meeting data privacy requirements for healthcare deployment.
- Built a semantic search pipeline using Sentence Transformers and FAISS, indexing 500+ structured Nigerian clinical protocols for sub-second retrieval at inference time.
- Integrated Whisper ASR and Tesseract OCR into a multimodal ingestion pipeline, enabling voice-driven and document-based query workflows for low-resource clinical environments.
- Engineered a token-efficient LLM prompting layer (Google Gemini) for grounded differential diagnosis generation, reducing lookup time for healthcare staff by 25%.

HabariPay – GTCO

Software Engineer Internship

Lagos, Nigeria (Hybrid)

Apr 2025 – Present

- Currently building enterprise events software for Squad, the fintech subsidiary arm of GTCO, supporting event management and payment workflows at scale.
- Built a full-stack vendor guide application (Next.js, Node.js, PostgreSQL) for a food festival, serving 260+ attendees and generating 2,100+ page views.
- Automated ingestion of vendor data from 2 sources into a centralized SQLite database, processing hundreds of records with high consistency.
- Implemented a TF-IDF recommendation engine and Gemini-powered summaries, improving user click-through rate by 7%.

Wema Bank

Data and AI Internship

Lagos, Nigeria (Hybrid)

Sep 2024 – Dec 2024

- Built a governance-compliant pipeline to extract and analyze 10,000+ Play Store reviews, producing structured product insights for cross-functional teams.
- Contributed to an AI-powered knowledge chatbot for Microsoft Teams (Node.js, Azure), enabling 2,000+ staff to query internal product and process documentation instantly.

NexaScale

Cloud Engineer Internship

Remote

Feb 2023 – May 2023

- Automated deployment of 30+ AWS resources with Terraform, reducing manual infrastructure setup by 75%.
- Built and deployed a microservice-based application with Node.js, containerized with Docker, and orchestrated on Kubernetes (AKS).
- Established a CI/CD pipeline using Azure DevOps, Git, and Docker for reliable, repeatable production deployments.

ZuriChat

Software Engineer & Lead Internship

Remote

Jul 2021 – Oct 2021

- Led a cross-functional team to ship real-time music and chess rooms using WebRTC, increasing user engagement by 20%.
- Improved backend performance by 50% through caching strategies, code refactoring, and API optimization.
- Managed multi-team deployments using GitHub Actions and Jenkins, ensuring stable integrations across concurrent feature branches.

PROJECTS

Virtue Foundation Intelligence Platform

Python, LangGraph, FAISS, GPT-4o-mini, FastAPI, Next.js

- Built an end-to-end agentic AI platform analyzing 797 Ghana healthcare facilities across 16 regions, identifying 10 medical deserts and 43 data anomalies through a multi-agent pipeline.

- Designed an Intelligent Document Processing engine (GPT-4o-mini) to extract structured capability data from unstructured facility records, with per-field confidence scoring (0-99%).
- Architected a LangGraph supervisor system routing natural language queries to 6 specialized sub-agents backed by a FAISS vector store on OpenAI embeddings; integrated ElevenLabs voice I/O for field accessibility.
- Won 2nd place at Databricks x Hack-Nation Global AI Hackathon.

Kinetix

Python, FastAPI, Next.js, TypeScript, Gemini API, WebSockets

- Built a real-time AI personal trainer using Gemini's vision API to analyze webcam exercise footage frame-by-frame, classifying posture and joint alignment as safe, suboptimal, or unsafe.
- Implemented WebSocket communication between FastAPI backend and Next.js frontend for low-latency feedback delivery; skeleton overlay with color-coded form indicators (green/yellow/red).
- Designed a dual-persona prompt architecture separating internal biomechanical reasoning from external coaching output, with post-session summaries and corrective exercise recommendations.

Modern Search Engine

C#, .NET, Azure, JavaScript

- Engineered a distributed information retrieval system with a custom inverted index, TF-IDF weighting, and cosine similarity, delivering sub-second query responses across 10,000+ PDF, DOCX, and TXT files.
- Designed and validated a robust text-processing pipeline (tokenization, stemming, stop-word removal) with comprehensive xUnit test coverage.

PheraCAM

Python, React.js, Azure, OpenAI API

- Developed a real-time face recognition system using OpenCV and Azure cloud services, achieving >95% accuracy with <150ms response time.
- Won 2nd place among 50+ teams at Hacklab's Hackathon, recognized for technical innovation and robustness.

AFCF Portal

Node.js, React.js

- Built a national reporting and coordination platform for 500+ users, automating compliance workflows and cutting manual data collation time by 75%.

RESEARCH & PUBLICATIONS

SharpXR: Structure-Aware Denoising for Pediatric Chest X-Rays

MIRASOL Workshop, MICCAI 2025 | [arXiv:2508.08518](#)

- Benchmarked seven denoising baselines; contributed to experimental evaluation and manuscript writing. SharpXR improved downstream pneumonia classification from 88.8% to 92.5%.

TECHNICAL SKILLS

- **Languages:** Python, Java, C#, JavaScript, TypeScript, SQL, Bash
- **Frameworks:** Node.js, Express, Next.js, Flask, FastAPI, React, Pandas, NumPy, OpenCV, PyTorch
- **Systems & Tools:** Docker, Kubernetes, Terraform, AWS, Azure, DigitalOcean, Git, Linux, CI/CD (GitHub Actions, Jenkins, Azure DevOps)
- **AI & Data:** LangGraph, FAISS, Hugging Face Transformers, Sentence Transformers, Scikit-learn, Whisper, OpenAI API, Gemini API

SCHOLARSHIPS

- Jim Ovia Undergraduate Scholarship (2025-2027): National scholarship recognizing top performance in STEM.
- Squad Hackademy Undergraduate Scholarship (2024-2027): Recognizing top performance in programming.
- PICFI Undergraduate Scholarship (2025-2027): National scholarship recognizing top performance in STEM.